



Submitted by

Name: Muhammad Abdul Kareem

Roll no: BSEF23M014

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Section: BS-SE (Morning)

Post Lab: CC-215-L DBS LAB-09 DDL

Subject: Database Systems

Task 6:

Post Lab

[15 Marks]

1. Create the following tables:

```
create table vendor  
( v_code CHAR(5) constraint vendor_id primary key,  
  v_name VARCHAR2(20) constraint vendor_name Not Null,  
  v_addr VARCHAR2(30)  
)
```

Table Name: vendor		
Col Name	Data Type	constraints
v_code	CHAR(5)	Primary Key
v_name	VARCHAR2(20)	Not Null
v_addr	VARCHAR2(30)	

```
create table purchase_invoice  
( p_invoice# NUMBER(5) constraint purc_invoice Primary Key,  
  p_invoice date DATE DEFAULT sysdate,  
  v_code char(5) Not Null,  
  constraint ven_id Foreign Key (v_code) references vendor (v_code)  
)
```

Table Name: purchase_invoice		
Col Name	Data Type	constraints
p_invoice#	NUMBER(5)	Primary Key
p_invoice date	DATE	DEFAULT sysdate
v_code	CHAR(5)	Foreign Key references vendor Not Null

```

create table purchase_invoice_details
(
  p_invoice# NUMBER(5),
  prod# NUMBER(5),
  qty_purchased NUMBER(4),
  CONSTRAINT purc_invo_detail_pk primary key (p_invoice#, prod#),
  CONSTRAINT purc_invo_fk foreign key (p_invoice#) REFERENCES purchase_invoice(p_invoice#),
  CONSTRAINT qty_purc CHECK (qty_purchased >= 0)
)

```

Table Name: purchase_invoice_details		
Col Name	Data Type	Constraints
p_invoice#	NUMBER(5)	Primary Key, Foreign Key references purchase_invoice
prod#	NUMBER(5)	Primary Key
qty_purchased	NUMBER(4)	Must not be negative

2. Populate the above tables with the following data.

```

INSERT ALL
  INTO VENDOR VALUES ('V001', 'ALI', '1234 Main Market')
  INTO VENDOR VALUES ('V002', 'HAMZA', '5678 Johar Town')
  INTO VENDOR VALUES ('V003', 'HASSAN', '91011 Garden Town')
SELECT * FROM dual;

```

select * from vendor

Results Explain Describe Saved SQL History

V_CODE	V_NAME	V_ADDR
V001	ALI	1234 Main Market
V002	HAMZA	5678 Johar Town
V003	HASSAN	91011 Garden Town

3 rows returned in 0.01 seconds [Download](#)

VENDOR

V_CODE	V_NAME	V_ADDR
V001	ALI	1234 Main Market
V002	HAMZA	5678 Johar Town
V003	HASSAN	91011 Garden Town

```

INSERT ALL
  INTO PURCHASE_INVOICE VALUES (101,to_date('May 2, 2025'),'V001')
  INTO PURCHASE_INVOICE VALUES (102,to_date('May 2, 2025'),'V001')
  INTO PURCHASE_INVOICE VALUES (103,to_date('May 2, 2025'),'V003')
SELECT * FROM dual;

```

```
select * from PURCHASE_INVOICE
```

Results Explain Describe Saved SQL History

P_INVOICE#	P_INVOICE_DATE	V_CODE
101	05/02/2025	V001
102	05/02/2025	V001
103	05/02/2025	V003

3 rows returned in 0.01 seconds [Download](#)

PURCHASE_INVOICE

p_invoice	p_inv_date	v_code
101	May 2, 2025	V001
102	May 2, 2025	V001
103	May 2, 2025	V003

```

INSERT ALL
  INTO PURCHASE_INVOICE_DETAILS VALUES (101, 21598, 10)
  INTO PURCHASE_INVOICE_DETAILS VALUES (101, 73956, 11)
  INTO PURCHASE_INVOICE_DETAILS VALUES (102, 16532, 50)
  INTO PURCHASE_INVOICE_DETAILS VALUES (102, 60482, 15)
  INTO PURCHASE_INVOICE_DETAILS VALUES (102, 86741, 22)
  INTO PURCHASE_INVOICE_DETAILS VALUES (103, 86741, 25)
  INTO PURCHASE_INVOICE_DETAILS VALUES (103, 60482, 16)
  INTO PURCHASE_INVOICE_DETAILS VALUES (103, 21598, 13)
  INTO PURCHASE_INVOICE_DETAILS VALUES (103, 73956, 10)
  INTO PURCHASE_INVOICE_DETAILS VALUES (103, 16532, 10)
SELECT * FROM dual

```

```
select * from PURCHASE_INVOICE_DETAILS
```

Results Explain Describe Saved SQL History

P_INVOICE#	PROD#	QTY_PURCHASED
101	21598	10
101	73956	11
102	16532	50
102	60482	15
102	86741	22
103	86741	25
103	60482	16
103	21598	13
103	73956	10
103	16532	10

10 rows returned in 0.00 seconds [Download](#)

PURCHASE_INVOICE_DETAILS

p_invoice#	prod#	qty_purchased
101	21598	10
101	73956	11
102	16532	50

102	60482	15
102	86741	22
103	86741	25
103	60482	16
103	21598	13
103	73956	10
103	16532	10